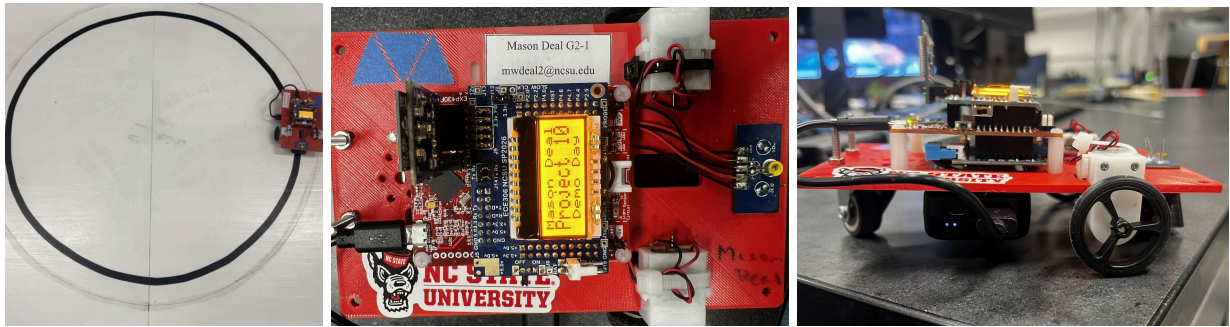


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Embedded Autonomous & WiFi Remote Controlled Car North Carolina State University - *Introduction to Embedded Systems*



Description:

Engineered a dual-mode autonomous and WiFi-enabled vehicle utilizing the Texas Instruments MSP430FR2355 microcontroller. Redesigned the power delivery system by migrating from AA batteries to a rechargeable lithium-ion solution, significantly enhancing system stability and operational uptime. Through extensive testing and adaptive PID control algorithms, the vehicle successfully completed a combined autonomous and Wi-Fi-controlled course on its first attempt.

Challenges faced:

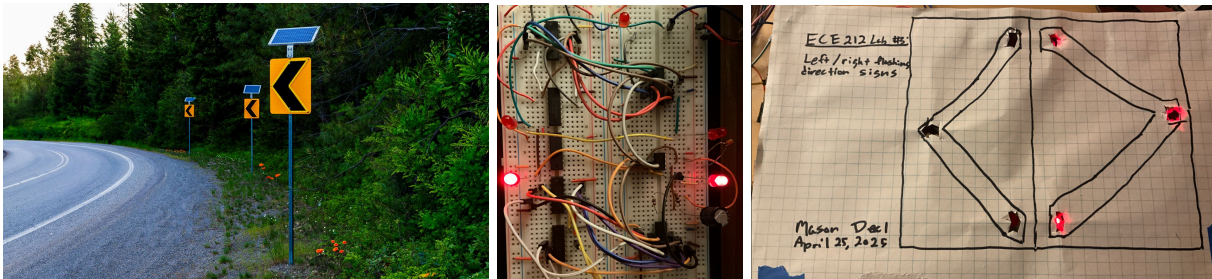
- Developed low-level firmware to manage peripheral integration, including hardware timers, interrupts, and serial IoT communication protocols.
- Managed complex hardware-software integration by interpreting schematics, debugging electrical systems, and working extensively with component datasheets.

How would I improve on this?

- Explore DAC-based voltage control for more precise speed regulation.
- Implement software-defined low-power sleep modes to improve energy efficiency and extend battery life.

LED Road Sign CMOS State Machine

North Carolina State University - *Introduction to Logic Design*



Description:

Designed and breadboarded a sequential logic circuit to emulate a curve-warning road sign, adhering to the full engineering design lifecycle from schematic design to physical demonstration.

Challenges faced:

- Designing and constructing two concurrent Finite State Machines (FSMs) to handle asynchronous user inputs and multi-pattern flashing sequences.
- Synchronized complex logic timing by troubleshooting and stabilizing the integrated clock circuit.

How would I improve on this?

- Integrate additional logic states to support synchronized "hazard" flashing modes.
- Optimize physical layout and wire routing to minimize interference and improve long-term reliability.